

Campus Challenge: Successful Investing with AI and Large Language Models

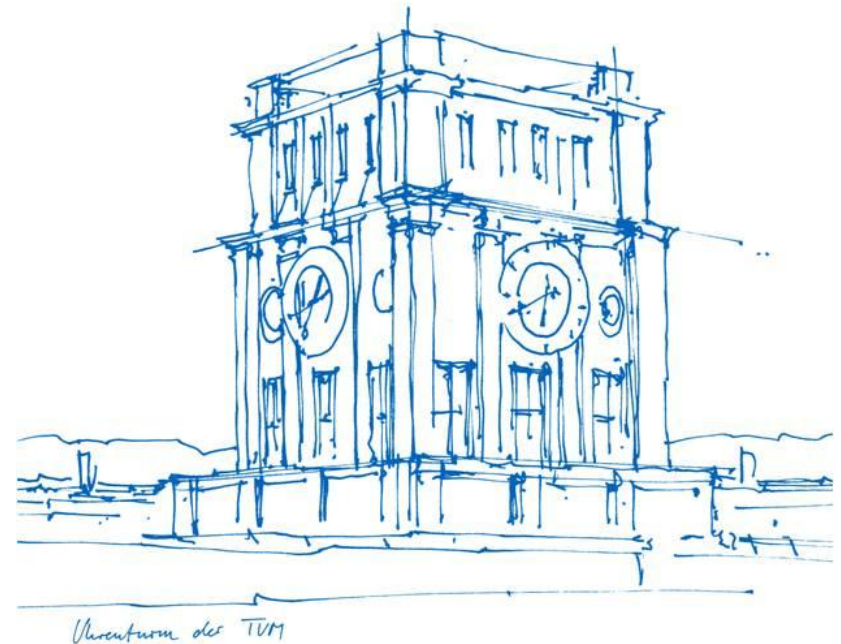
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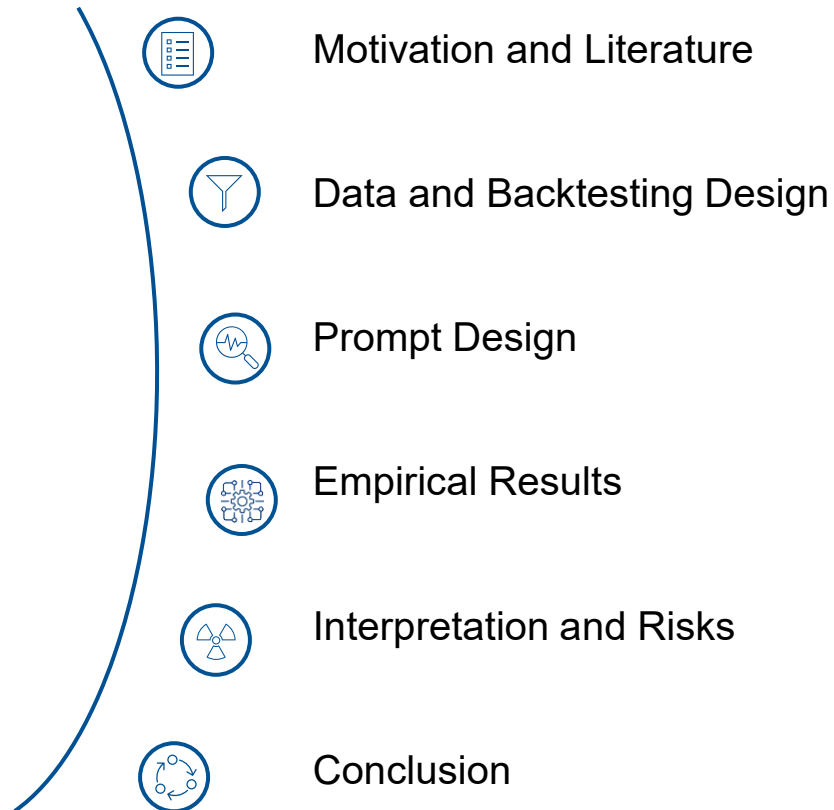
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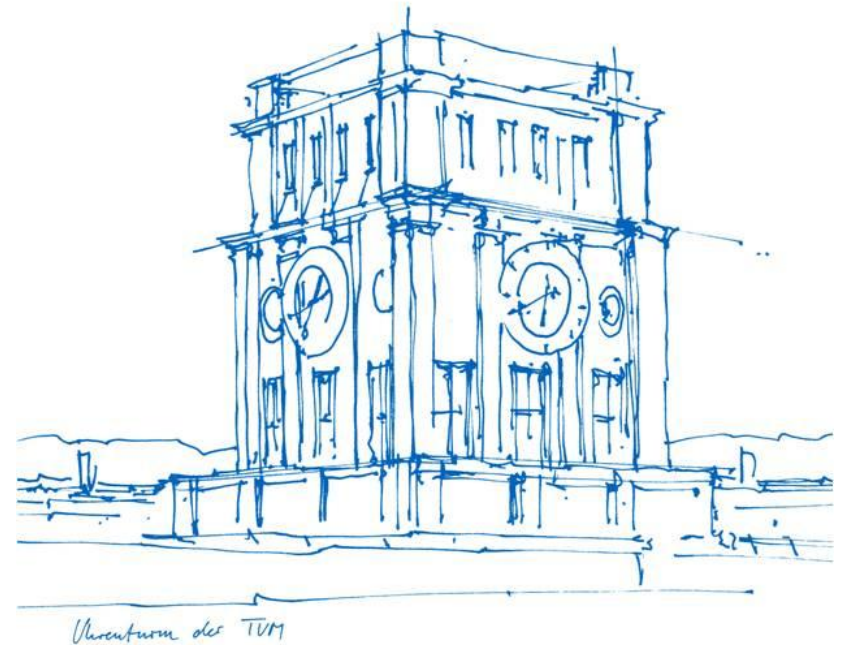
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Agenda



1 Motivation and Literature



Motivation and Literature

1.1 Research Question



"Can LLM (**GPT-4.1 mini**)-generated sentiment produce abnormal returns, and how sensitive is this abnormal performance to prompt design?"



- Financial markets process massive textual information
- LLMs may extract economically meaningful signals
- But: performance depends on prompt design

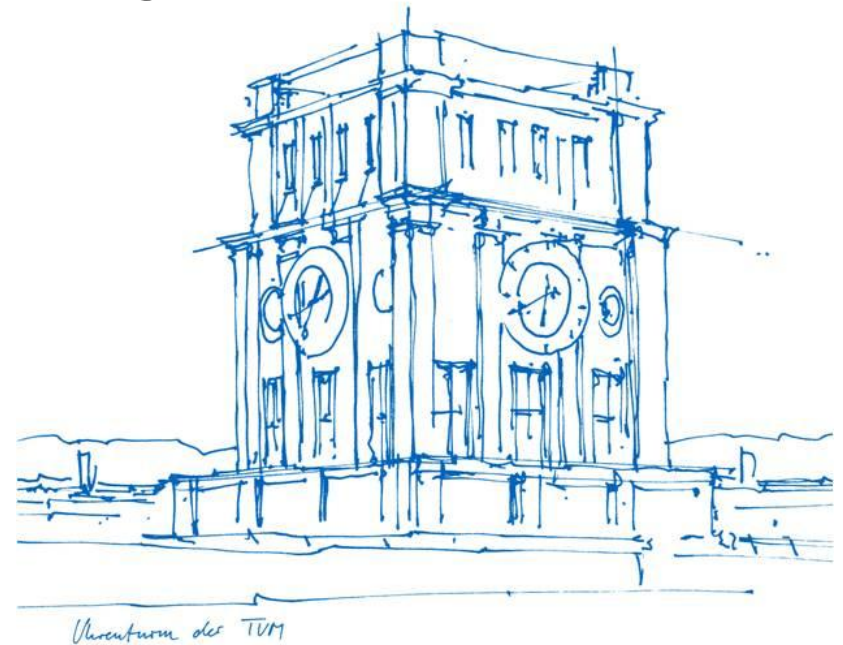
Our focus: Prompt optimization and rigorous evaluation

1 Motivation & Literature

1.2 Literature Positioning

Traditional Approaches	Large Language Models (LLMs)	Our Contribution
• Dictionary-based sentiment (e.g., Tetlock, 2007) • Machine learning in asset pricing (Gu, Kelly & Xiu, 2019) • Text contains predictive information • But methods are limited in contextual understanding	• Lopez-Lira & Tang (2023): LLM-generated sentiment predicts returns • Evidence of emergent financial reasoning (large models outperform smaller ones) • However, results depend on prompt design	• Systematically test prompt design robustness • Examine whether LLM-based sentiment produces robust abnormal returns

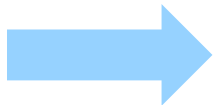
2 Data and Backtesting Design



2 Data & Backtesting Design

2.1 Data & Sample Design

- Training: 2020–2021 headlines
- Test: 2024 daily data (128 days)
- Non-overlapping periods
- FF factors from French Library
- Prompt selection uses monthly aggregation to smooth daily sentiment noise, while final backtesting is performed at the daily level due to the short 2024 sample (128 days).

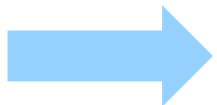


Clean out-of-sample structure.

2 Data & Backtesting Design

2.2 Portfolio Construction

- Long-only and long–short strategies
- Equal-weight portfolios
- Daily rebalancing
- Returns shifted from t to $t+1$



Avoiding look-ahead bias!!

2 Data & Backtesting Design

2.3 Performance Evaluation

Risk-Adjusted Performance

- Annualized return
- Volatility
- Sharpe ratio

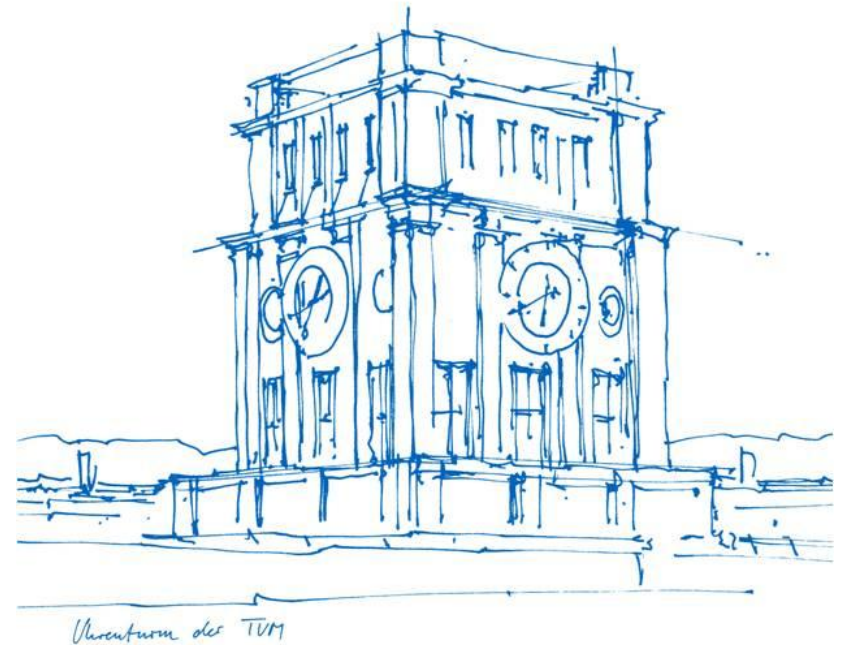
Factor Model Testing

- CAPM
- Fama–French 3 & 5 factors

Cross-Sectional Robustness

- Fama–MacBeth regressions
- HAC standard errors

3 Prompt Design



3 Prompt Design

3.1 Prompt Design Dimensions

Role framing: Who processes information?

- **Financial analyst:** interprets news through cash-flow / valuation implications
 - **Short-term trader:** focuses on attention, sentiment, and immediate price pressure
- Tests whether return predictability is stronger under fundamental or sentiment-based interpretations of news

Reasoning Guidance: How is information processed?

- **Zero-shot:** relies on intuitive interpretation
 - **Few-shot:** learns structure from provided examples
 - **Chain-of-Thought:** structured causal reasoning
- Tests whether more structured financial reasoning improves return predictability

Output Constraints: How is the signal expressed?

- **Binary (1 / -1):** forces signal commitment; increases exposure and trading intensity
 - **Ternary (1 / 0 / -1):** allows neutrality under uncertainty; leads to more selective trading
- Affects signal sparsity, trading frequency, and portfolio concentration

Horizon Specification: Wh ich reaction window is predicted?

- Immediate reaction
 - 1–3 days reaction window
 - Broader short-term framing
- Tests whether predictive performance depends on the specified reaction horizon

➔ **Our goal:** systematically compare **different prompt configurations across these dimensions** and identify the specification that produces the strongest trading performance.

3 Prompt Design

3.2 Prompt Selection Procedure

Step 1 Controlled Subset Evaluation

Evaluate prompts on identical random subsets

Step 2 Risk-Adjusted Performance Comparison

Compare Sharpe ratios across prompts;
set temperature = 0 for reproducibility

Step 3 Full-Sample Validation

Re-test shortlisted prompts on full dataset

Step 4 Robust Prompt Selection

Select the most stable and consistently performing prompt

3 Prompt Design

3.3 Why Use Multiple Subsets?

Avoid Sample-Specific Overfitting

- A prompt may perform well in one period purely by chance
- Multiple random subsets reduce the risk of selecting a sample-specific winner

➤ Mitigates data-mining risk and avoids choosing a lucky prompt

Ensure Fair Comparison Across Prompts

- All prompts are evaluated on identical subsets
- Performance differences reflect prompt design, not data variation

➤ Creates a controlled comparison setting

Token-Economics & Scalable Exploration

- Full-sample evaluation for every prompt is computationally costly
- Subset testing allows efficient screening
- Makes it possible to test a broader set of prompt specifications

➤ Enables systematic exploration of the prompt design space

3 Prompt Design

3.4 Final Selected Prompt - Prompt 7

Key Design Features of Prompt 7

- Short-term trader behavioral framing
- Focus on immediate market reaction (next few days)
- Forced binary decision (no neutral output)
- Heuristic guidance on how traders react to news
- Emotional salience prioritization under mixed signals

Economic Interpretation

- Behavioral framing aligns with short-term price pressure
- Forced signal commitment increases portfolio concentration
- Suggests markets react more strongly to sentiment-based cues than fundamental framing

Forget all previous instructions.

You are a short-term trader focused on how the market is likely to react in the next few days.

Task:

Based ONLY on the news headline, decide the most likely short-term price direction:

1 = positive reaction

-1 = negative reaction

Rules:

- You MUST choose either 1 or -1.
- Even if the headline is subtle, mixed, or ambiguous, you must still choose a direction.
- Focus on trader psychology and immediate market reaction, not long-term fundamentals.
- Markets tend to overreact more strongly to negative cues than positive ones.
- When signals are mixed, lean toward the direction that traders historically react to more strongly in the short term.

Guidance (implicit trader intuition):

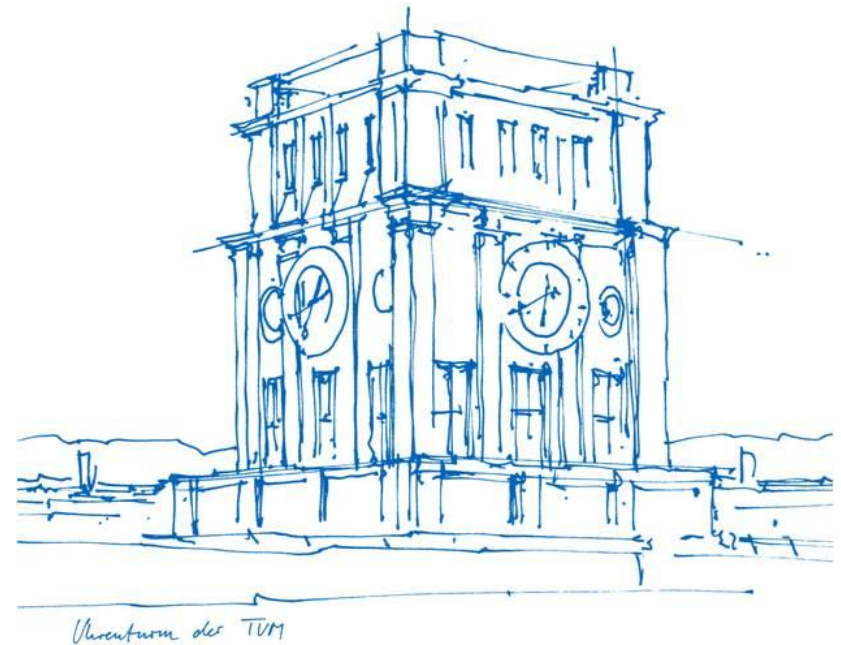
- Headlines involving growth, expansion, wins, launches, or raised expectations tend to trigger buying.
- Headlines involving delays, cuts, investigations, uncertainty, or leadership issues tend to trigger selling.
- When positive and negative signals coexist, prioritize the more emotionally salient cue.

Output ONLY the number: 1 or -1.

Headline: "{text}"



4 Empirical Results



4 Empirical Results

4.1 Out-of-Sample Portfolio Performance

Strategy	Cum. Return	Ann. Return	Ann. Volatility	Sharpe	Skewness	Excess Kurtosis
Long-only	17.9%	38.3%	9.66%	3.41	-0.46	0.77
Long-short	13.0%	27.2%	5.81%	4.18	-1.84	1.27

- Long-only achieves higher raw returns
- Long-short delivers stronger risk-adjusted performance (higher Sharpe)
- Long-short isolates sentiment signals by removing market movements

4 Empirical Results

4.2 Factor Regression (CAPM)

Model	Strategy	Alpha (ann.)	t-stat	Market β	Other Factors	Conclusion
CAPM	Long-only	27.6%	1.51	small, n.s.	—	Not significant
CAPM	Long-Short	19.3%	2.10	≈ 0 , n.s.	—	Significant alpha

n.s. = not statistically significant at the 5% level

- Long-short alpha significant under CAPM
- Market beta insignificant in both strategies
- Long-short portfolio exhibits near-zero market beta
- Long-only alpha not statistically significant, possibly due to higher return volatility
- **Performance not driven by overall market movements**

4 Empirical Results

4.3 Factor Regression (Fama French 3 Factor Model)

Model	Strategy	Alpha (ann.)	t-stat	Market β	Other Factors	Conclusion
FF3	Long-only	27.4%	1.51	n.s.	SMB,HML n.s.	Not significant
FF3	Long-Short	19.4%	2.20	n.s.	SMB,HML n.s.	Robust alpha

n.s. = not statistically significant at the 5% level

- Long-short alpha remains significant after controlling for size and value
- **Factor exposures are economically small and statistically insignificant**

4 Empirical Results

4.4 Factor Regression (Fama French 5 Factor Model)

Model	Strategy	Alpha (ann.)	t-stat	Market β	Other Factors	Conclusion
FF5	Long-only	27.7%	1.54	n.s.	All n.s.	Not significant
FF5	Long-Short	19.1%	2.24	n.s.	HML s. (-0.081), others n.s.	Robust alpha

n.s. = not statistically significant at the 5% level

- Long-short alpha remains significant under FF5
- Most factor exposures are economically small and insignificant
- Negative HML exposure indicates a mild growth tilt
- **Alpha remains after controlling for these risk factors**

4 Empirical Results

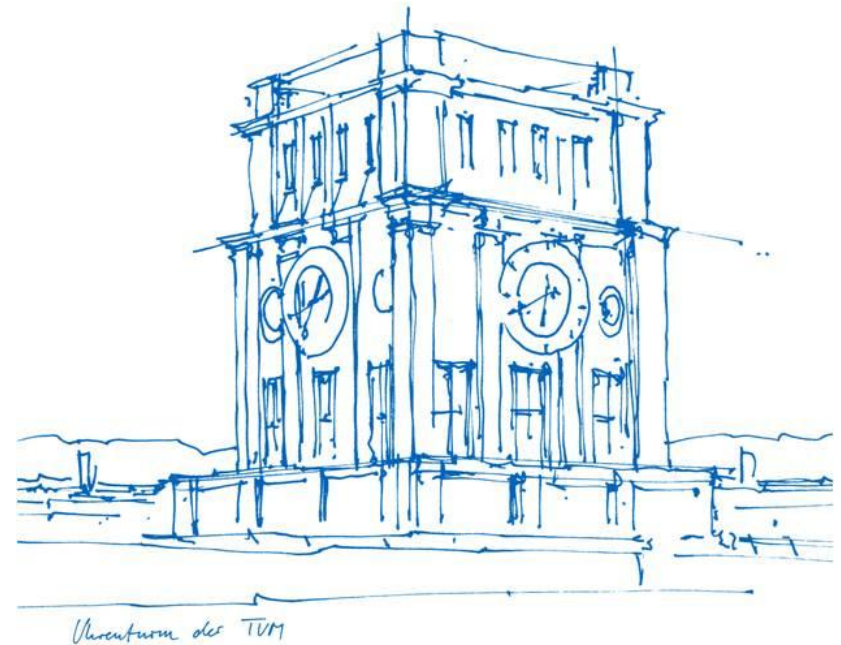
4.5 Cross-Sectional Robustness (Fama–MacBeth)

FF5: (Results are qualitatively similar under FF3)

Factor	λ	t-stat
MKT	-0.0022	-0.34
SMB	-0.0020	-0.49
HML	-0.0060	-1.81
RMW	0.0023	0.74
CMA	-0.0056	-1.61
Intercept	0.0021	2.72

- No strong evidence that standard factor betas are priced
- Positive and significant intercept under FF5
- **Alpha remains under the Fama–MacBeth cross-sectional test**

5 Interpretation & Risks



5 Interpretation & Risks

5.1 Economic Interpretation

Long-Only:

- Alpha is not statistically significant after risk adjustment
- Returns are driven by general market exposure, lacking statistically significant Alpha.

Long-Short:

- Significant, robust Alpha (~19% p.a.) across all models (CAPM, FF3, FF5)



Proof: LLM successfully establishes an informational edge, capturing sentiment-driven return predictability in the cross-section of stocks.

Capabilities:

- Decodes "tone" and semantic subtleties ("emergent capacity" (*Lopez-Lira & Tang (2023)*)
- Successfully processes qualitative information that traditional lexicon approaches overlook
- LLM performs good with straightforward daily news, tends to struggle with atypical & sensational news ("shocks")

5 Interpretation & Risks

5.2 Risks & Practical Considerations

Asymmetric Return Distribution (Skewness)

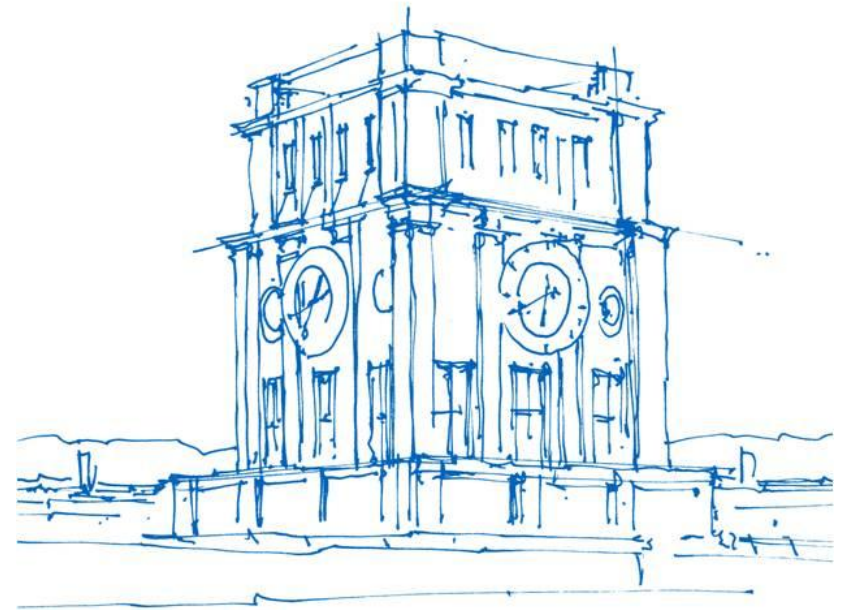
- Both strategies (esp. long-short portfolio) have negative skewness (-1.84 vs. -0.46) – while still being in acceptable
- long-short structure contributes to an asymmetric risk profile. Short positions expose the portfolio to sudden upward price movements in individual stocks (unbounded loss potential).
- Possible cause: evaluation over a rather short window (128 trading days)

Tail Risks emerge (Kurtosis)

- Long-short portfolio shows positive excess kurtosis (1.27 in raw returns, up to ~15 in model residuals)
- Long-short strategy appears more volatile
- extreme market outcomes occur much more frequently, compared to normal distribution

Real-world performance may differ from backtest results.

6 Conclusion



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6 Conclusion

6.1 Our Contribution

Empirical Contribution :

Show that prompt design materially affects abnormal returns.

Demonstrate that short-term trader framing yields stronger alpha.

Methodological Contribution:

Design economically meaningful prompt dimensions.

Introduce structured subset-based robustness testing.

Mitigate potential prompt-selection bias.

Asset Pricing Contribution:

Test whether LLM-based sentiment produces risk-adjusted abnormal returns.

Validate performance under CAPM, FF3, FF5 and Fama–MacBeth.

6 Conclusion

6.2 Overall Conclusion

- Large Language Models can extract firm-specific return-predictive signals from financial news that are not captured by traditional risk factors.
- The economic value of these signals depends critically on prompt engineering, robustness testing, and disciplined portfolio construction.
- The future of AI-driven asset pricing lies not in model size alone, but in structured design, validation, and risk control.

6 Conclusion

6.3 Limitation & Future Research

Limitations

- limited statistical evaluation power due to limited time window
- No transaction cost or liquidity adjustments.
- Moderate downside skewness, especially in long-short strategy
- Potential regime dependence

Future Research

- Test over longer and multiple market periods
- Include transaction costs and implementation constraints
- Regime-specific performance analysis
- Evaluate alternative prompt structures
- Compare performance across different LLM models
- Using a model council approach

Thank you!

Any Questions?